

Total No. of Questions : 6]

SEAT No. :

P3662

[Total No. of Pages : 2

APR - 15 / Engg. - 147
T.E. (IT) (In Sem - Semester - II)
SYSTEMS PROGRAMMING
(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Unit - I

- Q1)** a) Design a macroprocessor? Explain Nested Macro Call. **[6]**
- b) Show content of symbol table at the end of pass I and generate intermediate code. **[4]**

	START	100
A	DS	3
L1	MOVER	AREG,B
	ADD	AREG,C
	MOVEM	AREG,D
D	EQU	A+1
L2	PRINT	D
	ORIGIN	A-1
C	DC	'5'
	ORIGIN	L2+4
	STOP	
B	DC	'19'
	END	L1

OR

P.T.O.

- Q2)** a) Explain data structures of 2 pass assembler. [6]
b) What is language processor explains with diagram. [4]

Unit - II

- Q3)** a) What are the types of cards used in direct linking loader? Explain with example. [6]
b) Differentiate between absolute loader & relocatable loader. [4]

OR

- Q4)** a) Explain the concept of overlay with example. [4]
b) Explain compile and Go Loader scheme with its advantages and disadvantages. [6]

Unit - III

- Q5)** a) Write short note on LEX. [4]
b) Explain first three phases of compiler for given example (b2-4ac)/2a [6]

OR

- Q6)** a) Give regular expression/ definition for while space identifier and simple integer constant. Explain how they are used in recognizing the corresponding tokens. Support your answer with transition diagram. [4]
b) Convert RE to DFA for (a/b)*abb. [6]
